Polar Coordinates and Conversions

Section 1: Plotting with Positive and Negative Radii

1 How is a point with negative radius $(-r, \theta)$ plotted in polar coordinates?

a. Plot along the opposite ray directly across the pole (angle $\theta + \pi$)

b. Plot in the negative y direction

c. Reflect across the y-axis

d. The point is undefined

2 Which polar coordinate represents the same point as $(3, \pi/3)$?

a. $(-3, 4\pi/3)$

b. $(-3, 2\pi/3)$

c. $(3, 4\pi/3)$

d. $(-3, -\pi/3)$

3 In which quadrant does the polar point $(4, 7\pi/6)$ lie?

a. Quadrant III

b. Quadrant II

c. Quadrant IV

d. Quadrant I

Section 2: Converting Polar to Cartesian Coordinates

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 State the formulas to convert polar coordinates (r, θ) to Cartesian coordinates (x, y).

a. $x = r\cos\theta, y = r\sin\theta$

b. $x = r\sin\theta, y = r\cos\theta$

c. $x = r^2\cos\theta, y = r^2\sin\theta$

d. $x = \frac{\cos\theta}{r}, y = \frac{\sin\theta}{r}$

5 Convert the polar point $(6, 2\pi/3)$ to exact Cartesian coordinates.

a. $(-3, 3\sqrt{3})$

b. $(3, -3\sqrt{3})$

c. $(-3\sqrt{3}, 3)$

d. $(-3, -3\sqrt{3})$

6 Convert $(4, -\pi/4)$ to Cartesian coordinates.

a. $(2\sqrt{2}, -2\sqrt{2})$

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(2\sqrt{2}, 2\sqrt{2})$

d. $(4, -4)$

Section 3: Converting Cartesian to Polar Coordinates

7 Convert the Cartesian point $(-2, 2)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(2\sqrt{2}, 3\pi/4)$

b. $(2\sqrt{2}, 5\pi/4)$

c. $(4, 3\pi/4)$

d. $(2\sqrt{2}, \pi/4)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Convert $(0, -5)$ to polar coordinates.

a. $(5, 3\pi/2)$

b. $(5, \pi/2)$

c. $(-5, 0)$

d. $(5, \pi)$

9 Convert $(3, -3\sqrt{3})$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(6, 5\pi/3)$

b. $(6, 4\pi/3)$

c. $(6, 2\pi/3)$

d. $(12, 5\pi/3)$

Section 4: Converting Equations Between Polar and Cartesian

10 Convert the polar equation $r = 4\cos\theta$ to Cartesian form.

a. $(x-2)^2 + y^2 = 4$

b. $x^2 + y^2 = 4$

c. $x^2 + (y-2)^2 = 4$

d. $x = 4$

11 Convert $r = 5\csc\theta$ to Cartesian form.

a. $y = 5$

b. $x = 5$

c. $x^2 + y^2 = 25$

d. $y = 5x$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 Convert the line $y = 3x$ to a polar equation.

a. $\theta = \arctan(3)$

b. $r = 3$

c. $r = 3\cos\theta$

d. $\theta = 3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647